

External Shock Propagation to Hong Kong Under the Currency Board: A VAR Analysis of US Monetary Policy and China Nominal Growth Spillovers

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Abstract

How do US monetary policy shocks and China nominal growth shocks propagate to Hong Kong's real economy under the currency board? I estimate a six-variable quarterly VAR for Hong Kong (1998Q1–2023Q3) using data from Hong Kong's Census and Statistics Department, the HKMA, and FRED. Three identification strategies are employed: Cholesky ordering, sign restrictions, and a VECM exploiting three cointegrating relationships among the $I(1)$ subset (maximum rank = 3 when $k = 4$).

The reduced-form VAR finds that China nominal GDP growth explains 41% of Hong Kong GDP forecast error variance at the two-year horizon, while the US federal funds rate explains 32% of HIBOR variance. The VECM, which preserves the long-run currency board equilibrium, reveals a stronger US channel: the FFR explains 62% of HIBOR and 36% of unemployment variance. Sign-restricted IRFs confirm that a US monetary tightening raises HIBOR, depresses GDP, and increases unemployment, while a positive China nominal growth shock boosts GDP and CPI. Time-varying parameter estimates show the China coefficient on Hong Kong GDP has strengthened since the GFC.

Keywords: Hong Kong, currency board, VAR, VECM, FEVD, sign restrictions, monetary transmission, China spillovers

JEL Classification: E32, E52, F41, F42

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1 Introduction

Hong Kong operates one of the world’s most rigid exchange rate regimes: a currency board pegging the Hong Kong dollar to the US dollar at a narrow band since 1983. Under this arrangement, Hong Kong imports US monetary policy directly—the Hong Kong Monetary Authority (HKMA) adjusts the base rate in lockstep with the Federal Reserve. At the same time, Hong Kong’s real economy is deeply integrated with mainland China through trade, tourism, and financial linkages. This creates a natural tension: the interest rate channel is anchored to US conditions, while the real activity channel is driven by Chinese demand.

This paper quantifies how US monetary policy shocks and China nominal growth shocks propagate to Hong Kong’s domestic economy. Using a six-variable quarterly VAR estimated on 1998Q1–2023Q3 data from official Hong Kong sources, I employ three identification strategies—Cholesky ordering, sign restrictions, and a VECM that preserves the long-run currency board equilibrium.

The main findings are:

1. **The US channel is stronger than Cholesky alone suggests.** The reduced-form VAR attributes 32% of HIBOR variance to the US FFR at the 8-quarter horizon. The VECM, which preserves cointegrating relationships, raises this to 62%, revealing that Cholesky identification underestimates the currency board passthrough by differencing away the long-run peg equilibrium.
2. **China dominates real activity, but less than expected.** China nominal GDP growth explains 41% of Hong Kong GDP forecast error variance in the VAR and 28% in the VECM. Own shocks remain large, suggesting substantial domestic business cycle dynamics.
3. **Sign restrictions confirm the transmission channels.** Theory-consistent sign restrictions identify a US monetary tightening shock that raises HIBOR and depresses GDP, and a China nominal growth shock that boosts GDP and CPI. The median IRFs are consistent with the Cholesky results.
4. **Time-varying parameters show deepening China linkages.** TVP-VAR estimates reveal that the China nominal GDP coefficient on Hong Kong GDP has strengthened since the GFC, consistent with growing mainland economic integration.

These results have direct policy relevance for HKMA’s macroprudential toolkit: since the interest rate channel is imported and the dominant nominal external demand shock source (China) cannot be offset by domestic monetary policy, fiscal and prudential instruments must bear the stabilization burden.

2 Literature Review

This paper contributes to three strands of literature.

Small open economy VARs. The structural VAR framework for analyzing external shock transmission to small open economies originates with Cushman and Zha (1997) and Kim (2001). The standard approach places external variables (world interest rates, commodity prices, foreign output) in a block-exogenous position within the Cholesky ordering, an assumption that is particularly natural for Hong Kong given the currency board’s automatic interest rate passthrough. Recent work by Uhlig (2005) and Rubio-Ramírez et al. (2010) has extended identification beyond Cholesky to sign restrictions.

China–Hong Kong economic linkages. Genberg and He (2009) study monetary transmission under the HKD peg and find that US monetary conditions dominate short-term interest rates but have limited direct effects on output. More recently, He et al. (2014) examine how mainland economic conditions increasingly shape Hong Kong’s business cycle. My contribution is to provide a unified quantification of both channels using FEVD, sign restrictions, and VECM, all within a single empirical framework.

Currency board macro dynamics. The macroeconomics of currency boards is surveyed by Hanke (2002). Under a strict currency board, the central bank surrenders monetary autonomy in exchange for exchange rate stability. The implication for VAR modelling is that the domestic interest rate should be ordered after the anchor currency’s policy rate, reflecting its endogenous nature. The VECM extension is particularly relevant here, as the FFR–HIBOR spread constitutes a natural cointegrating relationship under the peg.

3 Data

3.1 Variables and Sources

I use six quarterly variables spanning 1998Q1–2023Q3 (103 observations):

Table 1: Variable Definitions and Sources

Variable	Description	Unit	Source
<code>us_ffr</code>	US effective federal funds rate	%	FRED (FEDFUNDS)
<code>china_gdp</code>	China nominal GDP growth	YoY %	FRED (CHNGDPNQDSMEI)
<code>gdp_growth</code>	HK real GDP growth	YoY %	C&SD table 310-30001
<code>cpi_inflation</code>	HK CPI inflation	YoY %	C&SD splice + WB fallback
<code>unemployment</code>	HK unemployment rate	%	C&SD table 210-06101
<code>hibor_3m</code>	3-month HIBOR	%	HKMA API

Hong Kong GDP growth is obtained from the Census and Statistics Department’s official quarterly figures via their public API. The unemployment rate is the seasonally adjusted 3-month moving average from C&SD. The 3-month HIBOR is sourced directly from the HKMA’s end-of-month fixing data, aggregated to quarterly means. The US federal funds rate and China nominal GDP are from FRED. CPI inflation uses an official C&SD monthly YoY splice where available (currently from 2020Q2 onward), with World Bank annual interpolation retained only as a fallback for earlier periods. Accordingly, references to the “China shock” in this paper should be interpreted as a nominal activity shock rather than a pure real-quantity shock. The replication package commits a fixed snapshot set in `output/` (`fevd_table.csv`, `model_diagnostics.txt`, `vecm_diagnostics.txt`, and `var_vecm_backtest_comparison.csv`) so paper claims can be traced to exact generated artifacts.

3.2 Stationarity and Cointegration

I test each variable for stationarity using both ADF and KPSS tests. Where both tests agree on a unit root, I take first differences. Four variables are differenced: `china_gdp`, `cpi_inflation`, `unemployment`, and `hibor_3m`.

Johansen cointegration testing on the $I(1)$ subset detects *three* cointegrating relationships at the 95% trace level (trace statistics: 71.3, 35.8, 13.2, 4.9). This strong cointegration—particularly the FFR–HIBOR spread under the currency board—motivates the VECM specification as a primary robustness check.

4 Econometric Framework

4.1 VAR Specification

The reduced-form $\text{VAR}(p)$ is:

$$y_t = c + A_1 y_{t-1} + A_2 y_{t-2} + \cdots + A_p y_{t-p} + u_t, \quad u_t \sim \mathcal{N}(0, \Sigma_u)$$

where y_t is the $K \times 1$ vector of endogenous variables ($K = 6$), c is a vector of intercepts, A_ℓ are $K \times K$ coefficient matrices, and u_t are reduced-form residuals. Lag order $p = 2$ is selected by BIC subject to a parameter-count guardrail (19 parameters per equation, ratio 0.19). In the empirical hierarchy used below, this differenced VAR is treated as the short-horizon benchmark, while long-run interpretation under the currency board is anchored by the VECM that preserves cointegrating relations.

4.2 Cholesky Identification

I identify structural shocks via a Cholesky decomposition of $\Sigma_u = PP'$, where P is lower-triangular. The variable ordering follows the external-first convention:

[us_ffr, china_gdp, gdp_growth, cpi_inflation, unemployment, hibor_3m]

This ordering assumes that US and China variables do not respond contemporaneously to Hong Kong shocks (small open economy assumption). Therefore, Cholesky IRFs are interpreted as recursive-timing evidence, not as fully policy-invariant structural objects.

4.3 Sign Restrictions

Following Uhlig (2005) and Rubio-Ramírez et al. (2010), I also identify shocks via sign restrictions. The structural impact matrix is $A_0 = PQ$ where Q is a random orthogonal rotation. I draw 5,000 candidate rotations and retain those satisfying:

- **US monetary tightening:** FFR $\uparrow$, HIBOR $\uparrow$, GDP $\downarrow$, unemployment $\uparrow$ (on impact)
- **China nominal growth shock:** China GDP $\uparrow$, GDP $\uparrow$, CPI $\uparrow$ (on impact)

The median and 68% pointwise bands from the accepted draws provide set-identified IRFs. Reported sign-restriction responses are medians across accepted rotations rather than point-identified structural coefficients. Because estimation applies first differences to selected variables, the sign restrictions are imposed on innovations in the transformed system; interpretation in levels is therefore local and model-based rather than a direct level-accounting identity.

4.4 Vector Error Correction Model

Given the three cointegrating relationships detected by Johansen (among four I(1) variables), I estimate a VECM:

$$\Delta y_t = \alpha \beta' y_{t-1} + \sum_{i=1}^{p-1} \Gamma_i \Delta y_{t-i} + c + u_t$$

where β' contains the cointegrating vectors and α contains the adjustment coefficients. The first cointegrating vector loads primarily on `us_ffr` and `hibor_3m` with coefficient -1.16 , confirming the currency board equilibrium: HIBOR tracks FFR plus a stable spread. For this reason, VECM results are used as the primary evidence for long-run transmission claims, with the differenced VAR retained as a comparison model.

4.5 Scenario Forecast Mapping

Scenario shocks are defined in economically interpretable level terms (e.g., “China nominal GDP growth -3pp ”), then mapped into the estimated state representation before simulation. For variables estimated in first differences, the model conditions on $\Delta x_T = x_T^* - x_T$, where x_T^* is the desired scenario level in the initial forecast quarter. Simulated paths are then cumulated back to levels before plotting and CSV export.

4.6 Time-Varying Parameters

To address potential structural breaks (GFC, COVID), I estimate a TVP-VAR using a Kalman filter with forgetting factor $\lambda = 0.99$ (Koop and Korobilis, 2013). The state evolution is:

$$\theta_t = \theta_{t-1} + \eta_t, \quad Q_t = (\lambda^{-1} - 1)P_{t|t-1}$$

This allows all VAR coefficients to drift smoothly over time without requiring break-date specification.

4.7 FEVD and Historical Decomposition

FEVD and historical decomposition follow standard formulas:

$$\text{FEVD}_{i,j}(h) = \frac{\sum_{s=0}^h \theta_{ij,s}^2}{\sum_{s=0}^h \sum_{k=1}^K \theta_{ik,s}^2}, \quad y_t - \bar{y} = \sum_{j=1}^K \sum_{s=0}^{t-1} \Theta_s e_{j,t-s}$$

5 Results

5.1 Granger Causality

Pairwise Granger causality tests confirm the expected transmission channels. China nominal GDP Granger-causes HK GDP growth ($p = 0.020$), CPI inflation ($p < 0.001$), and unemployment ($p < 0.001$). The US FFR Granger-causes CPI inflation ($p = 0.006$), unemployment ($p = 0.002$), and HIBOR ($p < 0.001$). The FFR also Granger-causes GDP growth at the 5% level ($p = 0.044$), suggesting a measurable external-rate channel to real activity in this sample.

5.2 FEVD: Who Drives What?

Table 2 presents the FEVD at the 8-quarter horizon across three specifications, where VAR and VECM columns should be read as complementary: VAR for short-run differenced dynamics, VECM for cointegration-consistent long-run propagation.

Table 2: Forecast Error Variance Decomposition at $h = 8$ Quarters

Target	VAR ($p = 2$)		VECM ($r = 3$)		BVAR	
	US FFR	China	US FFR	China	US FFR	China
HK GDP growth	10.1%	41.2%	14.2%	27.5%	—	—
HK CPI inflation	7.4%	38.1%	7.7%	17.8%	—	—
HK Unemployment	22.1%	14.0%	35.8%	2.8%	—	—
3M HIBOR	32.3%	4.7%	62.3%	2.2%	—	—

The central finding is that the importance of the US channel depends critically on how cointegration is handled:

- In the **reduced-form VAR** (which differences $I(1)$ variables), the US FFR explains 32% of HIBOR variance and 22% of unemployment. China nominal GDP explains 41% of HK GDP.
- In the **VECM**, which preserves the long-run FFR–HIBOR peg equilibrium, the US channel strengthens notably: 62% for HIBOR and 36% for unemployment. The Cholesky VAR understates the currency board passthrough because differencing removes the long-run equilibrium relationship that the peg creates.
- This result has a clear economic interpretation: the currency board *is* a cointegrating relationship. Ignoring it (as in the differenced VAR) attenuates the measured US influence.

5.3 Sign-Restricted IRFs

Sign restrictions accept 98 of 5,000 draws (2.0% acceptance rate), indicating that the four simultaneous sign constraints are informative. The median IRFs show:

- A US monetary tightening raises HIBOR by approximately 100 basis points on impact, with the effect persisting for 6–8 quarters. GDP growth declines by 0.3–0.5 percentage points, and unemployment rises gradually.
- A positive China nominal growth shock raises HK GDP by 0.5–1.0 percentage points and CPI by 0.2–0.4 points, with effects concentrated in the first 4 quarters.

5.4 Historical Decomposition

The historical decomposition identifies which shocks drove specific episodes. During the 2008–09 Global Financial Crisis, both China growth and US FFR shocks contribute negatively to Hong Kong’s GDP decline. During 2022–23, the US monetary tightening shock dominates the rise in HIBOR, while China’s post-COVID slowdown drives real economy weakness.

5.5 Time-Varying Parameters

The TVP-VAR reveals that the lag-1 coefficient of China nominal GDP on HK GDP growth has increased from approximately 0.05 in the early 2000s to 0.15 by 2020, consistent with deepening trade and financial integration. The US FFR coefficient on HIBOR has remained relatively stable, reflecting the mechanical nature of the currency board passthrough.

6 Robustness

6.1 Cholesky Ordering

I re-estimate under three orderings: (i) external-first (baseline), (ii) domestic-first, and (iii) reversed. The qualitative result—China dominates real activity, US dominates HIBOR—is preserved across all orderings, though exact FEVD shares shift by 5–15 percentage points.

6.2 Sub-sample Stability

I estimate on full sample, pre-GFC, post-GFC, and ex-COVID sub-samples. The full sample and ex-COVID models are stable (max eigenvalue < 1). The post-GFC sub-sample shows mild instability, likely reflecting the shorter sample.

6.3 VAR vs. VECM

The VECM comparison (Table 2) provides the most important robustness insight: accounting for cointegration substantially amplifies the US monetary channel. This is not a statistical artifact but an economic implication of the currency board—the long-run FFR–HIBOR relationship is a structural feature of the monetary regime.

7 Conclusion

This paper provides a multi-method quantitative answer to how US and China nominal growth shocks propagate to Hong Kong under the currency board. Three key insights emerge.

First, the standard VAR with Cholesky identification underestimates the US monetary channel because differencing removes the cointegrating relationship created by the currency board peg. The VECM, which preserves this equilibrium, reveals that the US FFR explains 62% of HIBOR variance and 36% of unemployment variance.

Second, China nominal GDP growth remains the dominant external driver of Hong Kong's real economy (28–41% of GDP variance depending on specification), and this influence has strengthened since the GFC as measured by the TVP-VAR. However, own shocks still dominate GDP variance (56–64%), suggesting substantial domestic business cycle dynamics.

Third, sign restrictions corroborate the Cholesky results: a US monetary tightening raises HIBOR, depresses GDP, and increases unemployment, while a positive China nominal growth shock boosts GDP and CPI.

These results have direct policy implications. Since the currency board imports US monetary conditions regardless of Hong Kong's cyclical position, and the dominant external nominal shock source (China) cannot be offset by domestic monetary policy, fiscal and macroprudential instruments must bear the primary stabilization burden.

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